CST8919
DevOps - Security and Compliance

Course Outline

2025-2026

Pre-requisite(s)	N/A
Co-requisite(s)	N/A
Prepared by	John Dallas
Approved by	John Dallas, Chair
Approval date	Thursday, December 18, 2025
Normative hours	42.00
Grading system	A+ Through F

Applicable Program(s)	Level	Core/Elective
1531X01FWO - Cloud Development and Operations	2	Core
1531X03FWO - Cloud Development and Operations	2	Core
1531X07PAO - Cloud Development and Operations	2	Core
1531X09FAO - Cloud Development and Operations	2	Core

Land Acknowledgement

Algonquin College campuses in Ottawa, Perth and Pembroke are located on the traditional unceded, and unsurrendered territory of the Anishinàbe Algonquin People. The Algonquin People have inhabited and cared for these lands since time immemorial. We take this time to express our gratitude and respect to them and to the land for all that it has provided and will continue to provide.

As a post-secondary institution, we acknowledge the harms done to Indigenous Peoples and are committed to learning from the past. We pledge to promote healing and resilience as we move forward in partnership with the Algonquin Nations, First Nations, Métis, and Inuit Peoples in a spirit of reconciliation.

While we recognize that territorial acknowledgements are only one step in cultivating greater respect and inclusion of Indigenous Peoples, we commit to accompanying these words with actions. We are dedicated to building a future and community that is better for all.

We pledge to continue exploring and making meaningful contributions to the Truth and Reconciliation Commission of Canada's Calls to Action.

Learn more about the College's commitment to Truth and Reconciliation: https://www.algonquincollege.com/tri

Course Description

Moving data and applications out of a self-hosted datacentre to a managed cloud provider raises many security and privacy concerns. Students implement robust security policies and the monitoring and compliance tools required to ensure those policies are correctly followed. Students use a combination of open source and cloud vendor-specific tools during lab simulations to design, deploy and test various security and governance scenarios.

Vocational Learning Outcomes

This course provides the opportunity for you to achieve the following outcomes:

1531X01FWO - Cloud Development and Operations

VLO 1 Create an appropriate architectural design to support the client`s secure virtual infrastructure and application development needs. (T, CP)

VLO 3 Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, A)

VLO 4 Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, A)

VLO 5 Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, A)

VLO 7 Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, CP)

1531X03FWO - Cloud Development and Operations

VLO 1 Create an appropriate architectural design to support the client`s secure virtual infrastructure and application development needs. (T, CP)

VLO 3 Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, A)

VLO 4 Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, A)

VLO 5 Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, A)

VLO 7 Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, CP)

1531X07PAO - Cloud Development and Operations

VLO 1 Create an appropriate architectural design to support the client`s secure virtual infrastructure and application development needs. (T, CP)

VLO 3 Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, A)

VLO 4 Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, A)

VLO 5 Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, A)

VLO 7 Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, CP)

1531X09FAO - Cloud Development and Operations

VLO 1 Create an appropriate architectural design to support the client`s secure virtual infrastructure and application development needs. (T, CP)

VLO 3 Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, A)

VLO 4 Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, A)

VLO 5 Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, A)

VLO 7 Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, CP)

Assessment Levels —T: Taught **A**: Assessed **CP**: Culminating Performance

Course Learning Outcomes / Elements of Performance

When you have earned credit for this course, you will have demonstrated the ability to:

1. Test various single-sign-on (SSO) and federated identity management solutions.(1,4)

- **1.1** Test different authentication protocols and standards.
- **1.2** Test the integration of various identity management solutions.
- **1.3** Configure SSO, federation, and identity provider configurations.

2. Analyze audit logs to identify potential security risks. (1,5,)

- **2.1** Describe various security protocols and systems, such as firewalls, intrusion detection systems (IDS), and encryption methods
- **2.2** State how these systems generate audit logs and the specific events they capture is crucial for interpreting audit data accurately.
- **2.3** Analyze logs using various log management solutions
- **2.4** Explain the structure of different log formats, including regular expressions.
- **2.5** Configure SIEM tools for log collection, parsing, and correlation rules
- **2.6** Create custom dashboards and alerts based on specific security criteria.
- **2.7** Conduct searches and queries within SIEM tools to extract meaningful insights from the vast amount of log data.

3. Use cloud platform tools to enforce and report on adherence to organizational policies and usage restrictions (1,7)

- **3.1** Implement policy monitoring tools to ensure that deployed solutions adhere to regulatory, governance, and legal requirements.
- **3.2** Conduct regular compliance assessments and audits to validate adherence to industry standards and legal frameworks.
- **3.3** Establish incident response procedures and conduct regular drills to ensure preparedness in case of security breaches.

4. Propose infrastructure configuration requirements that meet differing organizations' operational and security requirements. (5,7)

- **4.1** Utilize cloud platform consoles and tools effectively to configure security policies, access controls, and usage restrictions.
- **4.2** Translate compliance requirements into specific configurations within cloud platform tools to ensure adherence.
- **4.3** Design scalable and secure infrastructure solutions.
- **4.4** Conduct assessments and gather requirements for infrastructure configurations.

Learning Resources

Required Textbook:

- All required texts and resources will be provided in Brightspace.

Recommended Resources:

- All recommended texts and resources will be provided in Brightspace.

Hardware and Software:

- Cloud Service Provider (CSP) environment
- A computer machine (physical or virtual) with the following minimum system requirements: https://www.algonquincollege.com/byod/programs/better/
- Access to Source control such as GitHub

To access your course list, learning resources and costs, visit Booklist.
Please note that a learner must be registered in a specific course or program to view this information.
Program tuition and fees can be found on the Tuition and Fees Estimator for Full-time Programs page.

Learning Activities

During this course, you are likely to experience the following activities:

- Lectures and demonstrations
- Hybrid learning activities.
- Class discussions
- Presentations
- Assignments
- Quizzes
- Tests

Pre-defined Evaluation / Earning Credit

The following list provides evidence of this course's learning achievements and the outcomes they validate:

Assignment(s) (20%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Lab Activity(ies) (30%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Quiz(zes) (15%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Midterm Exam(s) (10%)

Validates Outcomes: CLO 1, CLO 2

Final Project (15%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Oral Exam (10%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Prior Learning Assessment and Recognition

Students who wish to apply for Prior Learning Assessment and Recognition (PLAR) need to demonstrate competency at a post-secondary level in all outlined course learning outcomes. Evidence of learning achievement for PLAR candidates includes:

- Portfolio
- Project/Assignment

Course Related Information

Grades
The minimum passing grade for the course is 50% (D-). Course evaluation is based on a variety of evaluated submissions including projects, quizzes, labs, assignments, oral exam, midterm and final exams; however, to pass the course, students must also achieve a minimum grade of 50% on the Oral Exam, regardless of their overall course average.

Program Information

1531X01FWO - Cloud Development and Operations

Cloud Usage
Students in this program may be exposed to Cloud Software through student accounts but they should never use their Credit Card for these services. Algonquin College will not be responsible for a students' utilization of Online Services or Cloud Based Applications.

Appropriate Use of Generative Artificial Intelligence (GenAI) Tools and AI-Assisted Resources
In alignment with Algonquin College Policy AA48 (Academic Integrity), students must follow the specific evaluation instructions for each assessment and consult the professor if they are unsure whether GenAI tools are permitted. For the purposes of this course, "GenAI/AI-assisted resources" includes any tool, device, software, browser feature, plug-in, or service that can generate, rewrite, summarize, translate, solve, or provide suggested answers or content (e.g., chatbots, writing/code assistants, AI search tools, and AI-enabled devices such as smart glasses or similar wearable technology). Unless an assessment explicitly allows such tools/resources, students should assume they are **not permitted**. Where use is allowed, students must clearly acknowledge and/or cite that use as directed by the instructor. Submitting AI-generated or AI-rewritten content as one's own work, using AI outside the permitted scope, or failing to acknowledge/cite permitted use may be treated as academic misconduct under AA48.

1531X03FWO - Cloud Development and Operations

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Department Information

STUDENT ACADEMIC RESPONSIBILITIES

Each student is responsible for:

- Knowing the due dates for marked out-of-class assignments.
- Attending all classes and knowing the dates of in-class marked assignments and exercises.
- Arriving to class on time and remaining for the duration of the scheduled class.
- Maintaining a folder of all work done in the course during the semester for validation claims in cases of disagreement with faculty.
- Keeping both paper and electronic copies of all assignments, marked and unmarked, in case papers are lost or go missing.
- Regularly checking both Brightspace announcements as well as one's Algonquin e-mail account for important messages from both professors and college administration.
- Participating in on-line and classroom exercises and activities as required.
- Retaining course outlines for possible future use to support applications for transfer of credit to other educational institutions.

Harassment/Discrimination/Violence will not be tolerated. Any form of harassment (sexual, racial, gender or disability-related), discrimination (direct or indirect), or violence, whether involving a professor and a student or amongst students, will not be tolerated on the college premises. Action taken will start with a formal warning and proceed to the full disciplinary actions as outlined in Algonquin College Policies - HR22 and SA07.
Harassment means one or a series of vexatious comment(s) (whether done verbally or through electronic means), or conduct related to one or more of the prohibited grounds that is known or ought reasonably to be known to be unwelcome/unwanted, offensive, intimidating, derogatory or hostile. This may include, but is not limited to: gestures, remarks, jokes, taunting, innuendo, display of offensive materials, offensive graffiti, threats, verbal or physical assault, stalking, slurs, shunning or exclusion related to the prohibited grounds.

For further information, a copy of the official policy statement can be obtained from the Student Association.

Violation of the Copyright Act
General – The Copyright Act makes it an offence to reproduce or distribute, in whatever format, any part of a publication without the prior written permission of the publisher. For complete details, see the Government of Canada website at http://laws.justice.gc.ca/en/C-42 . Make sure you give it due consideration, before deciding not to purchase a textbook or material required for your course.

Software Piracy - The Copyright Act has been updated to include software products. Be sure to carefully read the licensing agreement of any product you purchase or download, and understand the terms and conditions covering its use, installation and distribution (where applicable). Any infringement of licensing agreement makes you liable under the law.

Disruptive Behaviour is any conduct, or threatened conduct, that is disruptive to the learning process or that interferes with the well being of other members of the College community. It will not be tolerated. Members of the College community, both students and staff, have the right to learn and work in a secure and productive environment. The College will make every effort to protect that right. Incidents of disruptive behaviour must be reported in writing to the departmental Chair as quickly as possible. The Chair will hold a hearing to review available information and determine any sanctions that will be imposed. Disciplinary hearings can result in penalties ranging from a written warning to expulsion.

For further details, consult the Algonquin College Policies AA32, SA07 and IT01 in your Instaguide.

College Related Information

Algonquin College's policies have been developed to ensure the health, safety and security of all students, faculty and staff, and the proper and fair operation of the College as an academic institution and employer. Please refer to the Algonquin College Policies website for the most current policy information available at http://www.algonquincollege.com/policies/.

Students are especially encouraged to be aware of the following College expectations

Academic Integrity
Algonquin College is committed to the highest standards of academic integrity, and students are expected to uphold these standards throughout their learning journey with the College and post-graduation. Any academic work submitted by a student is expected to be their own work, unless designated otherwise, all human and technological assisted generated sources must be attributed. Refer to the College's Library web site for attribution support and resources. All students are expected to be familiar with the College's AA48: Academic Integrity policy which outlines student and staff roles and responsibilities, the process for addressing, determining and ruling final academic decisions regarding violations of academic integrity as well as what represents academic misconduct. Students with any questions about academic integrity course expectations including but not limited to use of technology, generative AI and other resources to support completion of coursework, are encouraged to speak to their professor and refer to the course weekly schedule, assessment instructions, course announcements and materials.

Centre for Accessible Learning (CAL)
Students with visible and/or non-visible disabilities are encouraged to register with the Centre for Accessible Learning (CAL) in order to be eligible for appropriate learning supports and/or accommodations. Students are strongly encouraged to make an appointment with the Centre for Accessible Learning as early as possible when starting a program. Once your needs are identified, a Letter of Accommodation (LOA) will be issued which you can share with your professors through the CAL Student Portal. If you are a returning student, please ensure that professors are given a copy of your LOA each semester.

Test Accommodations for Students Registered with CAL
Students who require testing accommodations will book their tests/exams through the CAL Student Portal. It is important to note the posted test/exam booking deadlines for the term. Test booking deadlines are communicated through CAL Disabilities Counsellors, CAL Testing Services, and other college wide communication. Additional information on the steps to book accommodated tests/exams with CAL Test Services can be found at Step-by-Step Overview.

Retroactive Accommodations
Students are expected to meet evaluation and completion deadlines as stated in course outline and weekly schedule documents. In circumstances where evaluation deadlines are missed or student performance has been affected by a temporary, significant and/or unanticipated change in functional ability related to disability (including mental health), interim or retroactive accommodations may be considered. For these instances it is advised you reach out to CAL or your professors as soon as possible. For other situations where deferral of evaluations may be warranted, please consult Algonquin College Policy AA21: Deferred Evaluation.

Network Credentials and College Email
Students at Algonquin College are provided with a college email account. This is the address that will be used when the College, your professors, or your fellow students communicate important information about your program or course activities. Your network credentials can be found in the ACSIS portal and you are expected to check your Algonquin email regularly and to use it to send and receive college-related email. Students are required to maintain the privacy of their AC login credentials. Sharing network credentials poses a security risk and is subject to disciplinary action. Your responsibility to protect your login credentials is governed by the Information Security Policy IT01. Support is available through the college Information Technology Service (ITS) at: https://www.algonquincollege.com/its/

Mental Health and Wellness Supports
Canada has launched a national Suicide Helpline at www.988.ca and a wide range of mental health and wellness resources for Algonquin students are available at www.algonquincollege.com/SSS/student-health-wellness.

Student Course Feedback
Algonquin College invites students to share their course experience by completing a student course feedback survey for each course they take. For further details consult Algonquin College Policy AA25: Student Course Feedback.

Use of Mobile Devices in Class
With the proliferation of small, personal mobile devices used for communications and data storage, Algonquin College believes there is a need to address their use during classes and examinations. During classes, the use of such devices unless authorized by your professor can be disruptive and disrespectful to others. During examinations, the use of such devices is generally prohibited unless authorized by your professor. Otherwise use is considered academic dishonesty in the form of cheating. For further details consult Algonquin College Policy AA32: Use of Electronic Devices in Class

Technology Requirements

Students are required to have access to a computer and to the internet. There may also be additional technology-related resources required to participate in a course that are not included in the course materials fee, such as headphones, webcams, specialized software, etc. Details on these requirements can be found in the Weekly Schedule document of the course for each course available on Brightspace.

Transfer of Credit

It is the student's responsibility to retain course outlines for possible future use to support applications for transfer of credit to other educational institutions.

Safe Harbour

In the event of an unexpected major event (pandemic, etc.), your course may have changes that are not reflected in the Course Outline. Should this happen, the Weekly Schedule document will have updated information about your course.